Earthquake signals and Kilauea eruptions

Predicting eruptive events at Pu'u O'o volcano, with a time-ordered evaluation

UE24CS352A Machine Learning
Mini-project write-up
Problem 64: Predicting Eruptive Events at Volcanoes

Prithvish Adhikari (PES1UG24AM206)
Piyush Shiv (PES1UG24AM191)

1 Problem statement

Volcano monitoring has long used earthquake activity as a warning sign. We ask two questions about Pu'u O'o on Kilauea's East Rift Zone, following the Stanford CS229 study by Mullet, Singh and Wong (PEEVED, 2019): **(1) classification** – is the volcano in an eruptive episode when a given earthquake occurs? **(2) regression** – how many days remain until the next eruption begins? The original study shuffled earthquakes into train and test sets. We keep its data and features but add a **time-ordered holdout** that scores only later earthquakes, which is the situation a real forecaster faces.

2 Dataset, labels and features

STAGE	EARTHQUAKES	WHAT HAPPENS
Raw ANSS catalog	6,485	6,485 events plus 49 eruption entries; no invalid rows or duplicates found
East Rift Zone	3,764	Geographic half-plane filter from the PEEVED notebook (same count as the paper)
Modeled	2,960	30-day feature warm-up, end of reliable eruption coverage (20 Jul 1986), known next onset
Erupting label	304 (10.3%)	Inside an episode's duration plus a 24-hour buffer – a strongly imbalanced target

Original seven features: longitude, latitude, depth, magnitude, and earthquake counts in the previous 1, 7 and 30 days. **Four added past-only features** (classification experiment): count in the previous 6 hours, time since the previous earthquake, mean and maximum magnitude over the previous 24 hours. Every count covers [t minus window, t), so the current event and all later events are excluded (data-leakage control, covered by unit tests). **Targets:** erupting (0/1) and days to the next eruption onset (0 if already erupting).

Our labels agree 100% with the original PEEVED rule on the modeled rows. Applied to all 3,764 East Rift events that rule marks 22.4% as erupting (paper: about 20%); our lower 10.3% comes from dropping 282 post-July-1986 events labelled only through episode 48's corrupted duration and 522 warm-up earthquakes of January 1983.

3 Approach and implementation

STEP	WHAT WE DO
Models (classification)	Prior baseline, logistic regression, random forest, Extra Trees, gradient boosting, class-conditional K-means, small MLP (32-16)
Models (regression)	Training-mean baseline, random forest, joint K-means, MLP
Time-ordered split	70% train / 20% development / 10% test by time, 30-day gaps, and labels that must be observable before the next window starts
Selection protocol	Scaling fit on training data only. Model, hyperparameters, feature set and decision threshold are chosen on development data; the test period is scored once
Comparison	A stratified random 70/20/10 split (as in the paper) is kept as a secondary experiment; code: src/data.py, models.py, experiment.py; demo.py scores a held-out quake

Data quality decisions. Episode 48 lists a duration of 173,566,800 hours (about 19,800 years): we keep its onset as the end of coverage and never use the duration. Eruptions 35.5 and 35 are out of date order: onsets and durations are sorted together.

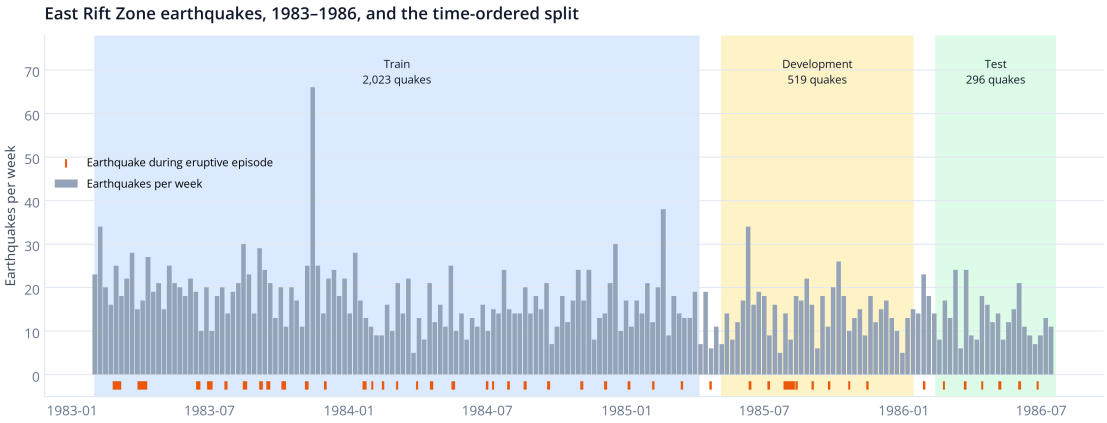


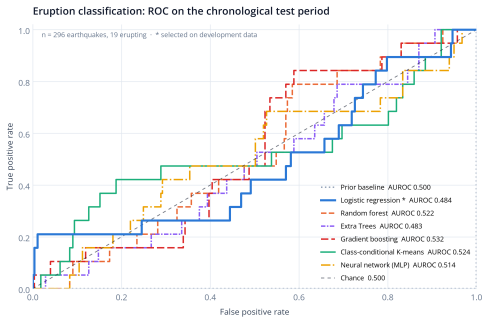
Figure 1. Weekly East Rift Zone earthquakes, eruptive-episode labels (orange ticks) and the train / development / test windows. Test holds 296 earthquakes, only 19 of them erupting.

4 Results

The chronological test period (Feb–Jul 1986) is the primary measure of performance on later earthquakes.

0.484	-0.006	7.75 d	-0.179
test AUROC	Cohen's kappa	regression RMSE	regression R^2

Classification: is the volcano erupting?



ROC curves on the chronological test period (n = 296 earthquakes, 19 erupting).

MODEL	TEST AUROC
Prior baseline	0.500
Logistic reg. (dev-F1 pick)	**0.484**
Random forest	0.522
Extra Trees	0.483
Gradient boosting	0.532
K-means (k=4)	0.524
MLP	0.514

Every model sits within chance range (0.48–0.53). With 19 positives the AUROC interval is wide (roughly 0.37–0.68 in an earlier bootstrap), so differences between models are not meaningful.

Development selected **logistic regression on the expanded 11-feature set** (C=10) with a development-F1 threshold of 0.611. On test it found **4 of 19** eruptive quakes (recall 0.211, precision 0.061, average precision 0.206, kappa -0.006) and flagged 62 repose quakes. At a fixed 0.5 threshold it finds 11 (F1 0.101). Selecting by AUROC alone would instead pick gradient boosting (development AUROC 0.603); the two procedures optimise different goals, so neither is automatically better.

OPERATING-POINT EXPERIMENT (TEST RESULTS)	THRESHOLD	FOUND / 19	NOTE
K-means k=4, fixed 0.5 (original)	0.500	12	194 false alarms; AUROC 0.524 for every row
K-means, development-F1 threshold	0.538	9	threshold changes decisions, not ranking
K-means, dev. FPR cap 0.10 / 0.20	dev-set	1 / 6	thresholds chosen on development data
Recall-first, FPR cap 0.10 (MLP)	0.458	1 (recall 0.053)	dev-selected; dev recall 0.152, test FPR 0.090
Recall-first, FPR cap 0.20 (logistic)	0.611	4 (recall 0.211)	dev-selected; test FPR 0.224 (not constrained)
Tuned K-means (k=3, relative distance)	dev-F1	3 (recall 0.158)	dev AUROC 0.668 but test 0.515, FPR 0.177

All thresholds, including the recall-first FPR caps, were selected on development data only; test labels were never used to choose them. The test FPR is reported but was not constrained.

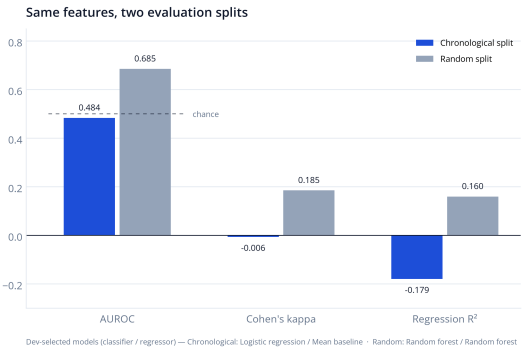
Key negative result. Searching K-means settings (k = 2 to 8, raw or scaled inputs, two distance scores) raised development AUROC to **0.668**, but the frozen chronological test AUROC was **0.515**, no better than the original k=4 model (0.524). The tuning improved development ranking only; it did not improve performance on later earthquakes.

Regression: days to the next eruption

REGRESSION MODEL	TEST RMSE (d)	MAE (d)	R^2
Mean baseline (selected)	**7.75**	**6.60**	**-0.179**
Random forest (depth 5)	7.30	6.25	-0.048
Joint K-means (k=10)	7.56	6.40	-0.121
MLP (alpha=0.1)	8.28	6.93	-0.345

Development chose the mean baseline. The forest has lower test RMSE, but it was worse on development data and switching after seeing test would be tuning on test. No model has positive test R^2.

Random vs time-ordered split



A random split lets quakes from the same swarm sit in both train and test, so it looks easier: random forest reaches AUROC 0.685, kappa 0.185 and regression R^2 0.160 (paper: AUROC 0.71, kappa 0.52, R^2 0.38 on a standardized target). We do not claim to reproduce those paper numbers.

5 Conclusion and limitations

With catalog features alone, models do not forecast eruptions reliably on later earthquakes: classification ranking is near chance and no regressor beats the mean. Tuning K-means improved development but not test AUROC. The main lesson is methodological: **evaluation design changes the story**, and leakage-aware time-ordered testing gives the honest answer. Limits: only 19 eruptive test events, 49 eruption entries in total, uncalibrated scores, and a single volcano. Next steps: ground deformation, gas emissions and focal mechanisms, and a longer catalog.

Sources: Mullet, Singh and Wong, "Predicting Eruptive Events at Volcanoes from Earthquake Data," Stanford CS229, Fall 2019; github.com/bmullet/PEEVED (ANSS catalog via WOVOdat; eruptions: Hawaii Center for Volcanology). All numbers are this project's own runs (scikit-learn 1.7.2).